

HKU Capstone Project Progress Update

Project: Deception-in-Depth Honeynet Architecture for Multi-Layer Cyber Deception

Current Progress Overview

Over the past several weeks, significant progress has been made in designing and implementing the foundational infrastructure for the Deception-in-Depth (DiD) honeynet environment. The project has successfully transitioned from conceptual planning into active infrastructure deployment and architectural validation.

The current implementation focuses on building a multi-layer deception environment capable of simulating realistic enterprise systems while supporting scalable remote management, monitoring, and future deception orchestration capabilities.

Achievements

1. Infrastructure Environment Successfully Established

A dedicated mini-server environment has been prepared to host the capstone infrastructure. Proxmox VE was selected as the virtualization platform due to its flexibility, snapshot capability, and suitability for multi-VM security experimentation.

The following components have been successfully configured:

- Proxmox hypervisor installation
- Virtual networking setup
- Initial VM provisioning workflow
- Snapshot strategy for rollback and recovery
- Ubuntu Server deployment baseline

This provides a stable foundation for future deception-layer deployment and experimentation.

2. Secure Remote Access Architecture Implemented

A secure remote-access model has been established using Tailscale to enable encrypted management access to the laboratory environment.

The implemented architecture now supports:

- Secure remote administration
- Remote troubleshooting
- Team collaboration access
- Reduced exposure of management interfaces to the public Internet

This significantly improves operational manageability while maintaining security isolation for the honeynet environment.

3. Initial Deception Layer Planning Completed

The project has completed the initial architectural planning for the Deception-in-Depth model.

The current design separates the environment into multiple deception layers, including:

- Infrastructure layer
- Network/service deception layer
- Interactive deception components
- Future orchestration and trigger-response mechanisms

The team also evaluated the integration potential of T-Pot as the primary honeypot framework for the network deception layer.

4. Documentation and Knowledge Management Established

A structured Notion workspace has been developed to centralize project documentation, deployment procedures, architecture diagrams, and operational notes.

The handbook currently includes:

- deployment procedures
- network architecture notes
- remote access documentation
- infrastructure setup steps
- artifact tracking concepts
- troubleshooting observations

This documentation improves reproducibility and supports future maintenance and demonstration activities.

Challenges Faced

1. Infrastructure and Network Configuration Complexity

One major challenge encountered was the complexity of configuring virtualization networking within Proxmox. Issues related to bridge interfaces, IP assignment, and connectivity troubleshooting required multiple iterations before stable communication between components was achieved.

Additional setup issues included:

- package repository synchronization
- Linux networking configuration
- remote connectivity troubleshooting
- VM deployment consistency

These issues increased setup time but provided valuable operational experience in infrastructure administration.

2. Balancing Realism and Safety in Honeynet Design

Another challenge is designing a deception environment that appears realistic enough to attract and engage attackers while ensuring the environment remains isolated and safe from unintended misuse.

The project therefore requires careful consideration of:

- network segmentation
- containment controls
- service exposure
- logging visibility
- operational monitoring

This remains an active design consideration moving into the next phase.

3. Time Allocation Between Research and Implementation

The project involves both academic research and practical system implementation. Balancing literature review, architectural design, infrastructure deployment, and documentation simultaneously has been challenging.

As the system becomes more complex, project coordination and prioritization become increasingly important.

Plans for Upcoming Weeks

1. T-Pot Deployment and Validation

The next major milestone is the deployment and stabilization of the T-Pot honeypot framework within the Proxmox environment.

Planned tasks include:

- deploying the T-Pot VM
- validating sensor functionality
- configuring logging and dashboards
- testing network visibility
- evaluating resource utilization

2. Development of Multi-Layer Deception Workflow

The project will begin implementing deeper deception interactions between layers.

This includes exploring:

- trigger-based deception escalation
- attacker interaction tracking
- layered engagement concepts
- centralized monitoring logic

The objective is to move beyond standalone honeypots toward a coordinated deception architecture.

3. Artifact Registry and Monitoring Expansion

The Notion-based artifact registry will be expanded to track:

- deployed services
- credentials
- interaction events
- deception assets
- attacker activity observations

This will support future analysis and reporting.

4. Research and Evaluation Preparation

The upcoming phase will also focus on defining evaluation methodology for the capstone project.

Potential evaluation areas include:

- deployment reproducibility
- deception realism
- operational stability
- monitoring effectiveness
- attacker interaction visibility

These metrics will later support the final report and demonstration.

Overall Status

The project is currently progressing according to its planned direction. The foundational infrastructure and remote management architecture have already been established, and the project is now transitioning into the implementation and experimentation phase of the deception environment.

Despite several infrastructure and configuration challenges, the project has successfully achieved a stable technical baseline that supports continued development in the coming weeks.